University of Florida, Department of Mathematics

Algebra PhD Exam, January 2003

Do seven of the following eleven problems. Please do not turn in more than seven problems. You must show your work. Answers with no work and/or no explanations will receive no credit. State clearly any theorem you use in your proofs. In the problems, $\mathbb{Z}$, resp. $\mathbb{Q}$, $\mathbb{C}$, is the set of all integers, resp. of all rational numbers, of all complex numbers.

1. State and prove the Jordan–Hölder Theorem for finite groups.

2. Let F be the free group in two generators a and b. Let N be the smallest normal subgroup of F which contains a^5, b^2 and $abab$. Prove that F/N is a non-abelian group of order 10.

3. Let G be an abelian group. Define the torsion subgroup G_t of G. Prove from your definition that G_t is a subgroup of G and that G/G_t is torsion free.

4. Let ζ be a primitive 2003-rd root of unity. Prove that $\mathbb{Q}(\zeta)$ is a Galois extension of $\mathbb{Q}$. Calculate the Galois group of $\mathbb{Q}(\zeta)/\mathbb{Q}$ and the degree $[\mathbb{Q}(\zeta):\mathbb{Q}]$ of the extension. Furthermore, calculate the degree $[\mathbb{Q}(\zeta)\cap\mathbb{R}:\mathbb{Q}]$. (Hint: 2003 is a prime).

5. A commutative ring R with identity is said to be Noetherian if every ideal of R is finitely generated. Let R be a Noetherian ring, and let C be an ascending chain of ideals of R. Prove that C only contains a finite number of distinct ideals.

6. Prove the following version of the Lying-Over Theorem. Let S be an integral extension ring of a commutative ring R with identity. Let P be a prime ideal of R. Prove that there exists some prime ideal Q of S such that $Q\cap R = P$.

7. Let $A=\mathbb{Z}/2\mathbb{Z}$ and $B=\mathbb{Z}/3\mathbb{Z}$ be viewed as $\mathbb{Z}$-modules. Calculate the $\mathbb{Z}$-module $A\otimes B$, and, in particular, give the cardinality of $A\otimes B$.

8. Among the following integral domains, decide which ones are Dedekind domains, and give a brief explanation.

(a) $\mathbb{Z}[T]$, the polynomial ring over the integers, in one variable.

(b) $\mathbb{Z}[\sqrt{-5}]=\{a+b\sqrt{-5} : a,b\in\mathbb{Z}\}$.

(c) The ring $k[[T]]$ of all formal power series in one variable, over the field k.

(d) $k[T_1,T_2]$, the polynomial ring in two variables, over the field k.

9. Give one representative for each conjugacy class of elements of order 4 in $GL(4,\mathbb{Q})$.

10. Calculate the number of irreducible polynomials of degree 4 over the field of 2 elements.

11. Let M be a semi-simple module over a ring R. Prove that every homomorphic image of M is semi-simple.